Cardano's Cubic

The equation that forced mathematics to take $\sqrt{-1}$ seriously

Where roots of polynomial equations meet the arithmetic of complex numbers

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Colour key: green = verification check | blue = alternative method.

The historical example is

$$\boxed{x^3 = 15x + 4} \qquad \text{equivalently} \qquad x^3 - 15x - 4 = 0.$$

It looks tame — and it is. Yet when the sixteenth-century formula for solving cubics is applied to it, a square root of a *negative* number appears, even though every root of the equation is real. Resolving that contradiction is what first made imaginary numbers respectable.

1. An honest, real equation

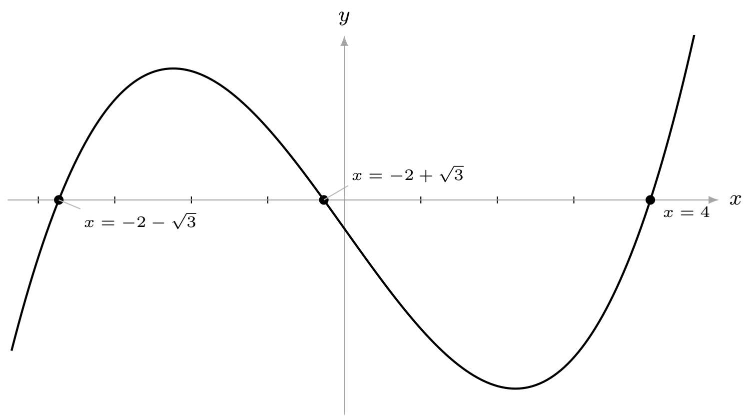
First, solve it the modern way. By inspection $x = 4$ works $(64 = 60 + 4)$, so $(x - 4)$ is a factor:

$$x^3 - 15x - 4 = (x-4)\big(x^2 + 4x + 1\big).$$

The quadratic gives $x = \dfrac{-4 \pm \sqrt{16-4}}{2} = -2 \pm \sqrt{3}$. Hence the three roots are

$$\boxed{x = 4, \qquad x = -2 + \sqrt{3} \approx -0.268, \qquad x = -2 - \sqrt{3} \approx -3.732}$$

— all three perfectly **real and distinct**. The cubic crosses the x-axis three times:



Check. Sum of roots $= 4 + (-2+\sqrt{3}) + (-2-\sqrt{3}) = 0$, matching the absent x^2 term ✓.
Product $= 4(-2+\sqrt{3})(-2-\sqrt{3}) = 4(4-3) = 4 = -\dfrac{(-4)}{1}$ ✓.

2. Cardano's formula, and where it breaks

The equation is already *depressed* (no x^2 term), so it has the form $x^3 + px + q = 0$ with $p = -15$, $q = -4$. Cardano's formula (from the *Ars Magna*, 1545) gives one root as

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}}, \qquad \Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3.$$

Substituting the numbers:

$$\Delta = (-2)^2 + (-5)^3 = 4 - 125 = -121, \qquad \text{so} \qquad \sqrt{\Delta} = \sqrt{-121} = 11i.$$

The formula therefore demands

$$x = \sqrt[3]{2+11i} + \sqrt[3]{2-11i}\,.$$

A real, well-behaved equation has driven us straight into $\sqrt{-1}$. To Cardano's contemporaries this looked like nonsense, yet the left-hand side is genuinely 4.

Check. For a cubic with three distinct real roots the quantity Δ is always negative. Here $\Delta = -121 < 0$ ✓, so the appearance of an imaginary square root is not a blunder — it is forced by the geometry of three real crossings.

3. Bombelli's leap of faith

In 1572 Rafael Bombelli made the bold move of *calculating with* $\sqrt{-1}$ as if it obeyed the ordinary rules of algebra. He guessed that the two cube roots were themselves of the form $2 \pm i$, and checked it directly:

$$(2+i)^3 = 2^3 + 3(2^2)i + 3(2)i^2 + i^3 = 8 + 12i - 6 - i = 2 + 11i.$$

Taking conjugates gives $(2-i)^3 = 2 - 11i$ at once. Hence

$$\sqrt[3]{2+11i} = 2+i, \qquad \sqrt[3]{2-11i} = 2-i,$$

and Cardano's expression collapses beautifully:

$$x = (2+i) + (2-i) = \boxed{4}.$$

The imaginary parts are equal and opposite, so they cancel — leaving the honest real answer. The "impossible" numbers were only ever scaffolding; they vanish from the final result.

Check. $(2+i)$ and $(2-i)$ are a conjugate pair, and any number plus its conjugate is real: $(a+bi)+(a-bi) = 2a$. Here $2a = 4$ ✓.

4. Recovering all three roots with de Moivre

Bombelli's pairing gives only $x = 4$. The other two real roots are hiding in the *other* cube roots — and this is where the example calls on complex numbers in earnest, through de Moivre's theorem and the Argand diagram. Write $2 + 11i$ in modulus–argument form:

$$|2+11i| = \sqrt{2^2+11^2} = \sqrt{125} = 5\sqrt{5} = \left(\sqrt{5}\right)^3, \qquad \theta = \arctan\frac{11}{2} \approx 1.391 \text{ rad}.$$

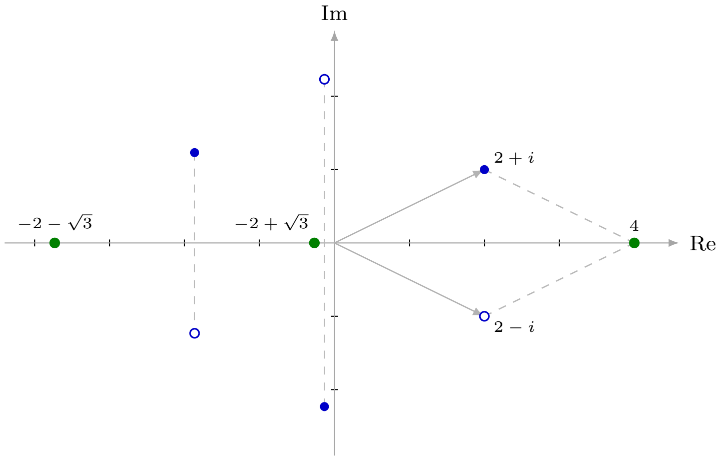
By de Moivre's theorem each cube root has modulus $\sqrt{5}$ and argument $\frac{\theta}{3} + \frac{2\pi k}{3}$ for $k = 0, 1, 2$:

$$w_0 = 2+i, \qquad w_1 \approx -1.866 + 1.232i, \qquad w_2 \approx -0.134 - 2.232i.$$

The cube roots of $2 - 11i$ are simply the conjugates $\overline{w_0}, \overline{w_1}, \overline{w_2}$. Pairing each root with its conjugate (so the imaginary parts cancel) gives three real sums:

$$w_0 + \overline{w_0} = 4, \qquad w_1 + \overline{w_1} \approx -3.732, \qquad w_2 + \overline{w_2} \approx -0.268,$$

which are exactly 4, $-2-\sqrt{3}$, $-2+\sqrt{3}$ from Section 1. Every real root is the sum of a conjugate pair of cube roots:



Filled dots: cube roots of $2+11i$; open dots: their conjugates (cube roots of $2-11i$). For the principal pair the grey arrows show $2+i$ and $2-i$ adding (parallelogram rule) to the green root 4; each conjugate pair likewise sums to a green real root as its imaginary parts cancel.

Check. Cube-root modulus $\sqrt{5}$ cubed is $5\sqrt{5} = |2{+}11i|$ ✓, and the three real sums $4,\ -3.732,\ -0.268$ agree with $4,\ -2 \mp \sqrt{3}$ to three decimals ✓.

5. The casus irreducibilis

This example is not a curiosity — it is a whole class of cubics. Whenever a cubic has **three distinct real roots**, the discriminant quantity satisfies $\Delta < 0$, so Cardano's formula *must* pass through $\sqrt{-1}$. This is the famous *casus irreducibilis* ("the irreducible case"). It was later proved that in this situation the real roots genuinely *cannot* be written using only real radicals: the detour through complex numbers is unavoidable, not merely convenient.

There is, however, a fully real — if less elementary — route, using trigonometry rather than radicals.

Alternative — Viète's trigonometric solution. For $x^3 + px + q = 0$ in the irreducible case ($p < 0$, $\Delta < 0$), the three real roots are

$$x_k = 2\sqrt{-\frac{p}{3}}\,\cos\left[\frac{1}{3}\arccos\left(\frac{3q}{2p}\sqrt{\frac{-3}{p}}\right) - \frac{2\pi k}{3}\right], \qquad k = 0, 1, 2.$$

With $p = -15,\ q = -4$: $2\sqrt{5} \approx 4.472$ and the inner arccos argument is 0.1789, giving

$$x_0 = 4.000, \qquad x_1 = -0.268, \qquad x_2 = -3.732,$$

exactly the three roots — obtained without ever writing $\sqrt{-1}$. The cube roots of a complex number have quietly become cosines of a trisected angle.

Check. The trig formula reproduces all three roots to three decimals ✓, and the cube-root arguments $\frac{\theta}{3}$, $\frac{\theta}{3} \pm 120^\circ$ of Section 4 are the same angle trisection appearing inside the arccos here.

6. Why it matters

Bombelli could not say what $\sqrt{-1}$ "was," but he could compute with it and watch it deliver correct real answers. That pragmatic success is what kept complex numbers alive for the two centuries before Euler, Argand and Gauss gave them a secure footing. The lesson endures: complex numbers are not an exotic add-on but the natural arithmetic in which real polynomial problems are most cleanly solved — here, three honest real roots that can only be reached by passing through the imaginary plane and back.

Note on attribution: the formula is Cardano's (1545); the $x^3 = 15x + 4$ *example and the imaginary-number arithmetic that resolves it are due to Bombelli (1572). Cardano's own encounter with* $\sqrt{-1}$ *was the problem of splitting* 10 *into two parts of product* 40*, giving* $5 \pm \sqrt{-15}$.